

Estimating Pitching Airfoil Kinematics Using Convolutional Neural Networks

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Convolutional neural networks(CNNs) have recently proven to be powerful in analyzing fluid flows. We aim to propose a novel CNN approach to estimate the airfoil pitching kinematics from a small, instantaneous snapshot from the downstream of the unsteady wake. We first focused on a sinusoidal pitching airfoil, which was created from DNS. The prediction of the airfoil base angle of attack and pitching frequency could be done with extremely high accuracy. Therefore, we further challenged the model by reducing the size of the input image in two ways: 1. reducing the measuring field area and 2. downsampling the image. In the former way of reduction, it was reduced until the field edge length was $0.8 \times$ chord length. Even in this substantial reduction, the accuracy scored 90.631%. In the later way of reduction, downsampling preserved the accuracy very well, making only 0.215% accuracy drop even in the most extreme case. We found that the window size of velocity measurement dominated the prediction accuracy in CNNs. The model was further challenged by using chirp signal pitching prediction, which was created by DNS as well. We first tested the performance of chirp signal prediction on models trained from sinusoidal pitching data. This was not feasible, as the model could not find the commonness between flows of the same base angle of attack and frequency when the frequency was changing. However, when the model was fed with varying frequency training data, the model could predict the kinematics with high accuracy (95.5%), while performing well on sinusoidal cases as well (99.8%). However, the model was found to be lacking in robustness when encountering an out-of-distribution data again for the prediction of chirp signal data that had different rates of frequency increase.

Nomenclature

- α_0 = the base angle of attack in oscillation ($^\circ$)
 α = the instantaneous angle of attack ($^\circ$)
 f_p = dimensionless frequency (constant)

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ω = instantaneous dimensionless frequency (varying frequency)
 c = chord length
 U_∞ = freestream velocity
 α_p = pitching amplitude
 U_x = Streamwise velocity (x -velocity)
 U_y = Wall-normal velocity (y -velocity)

I. Introduction

RAPID motion in aerial vehicles can cause substantial wake fluctuations. The affected wake behind the airfoil becomes very unsteady. In addition to rapid motions, aeroelastic flutter or flapping wings may also cause the wake to be unsteady. It is important to understand the unsteady aerodynamics to control any system within or influenced by the wake. This is challenging as there are too many varying parameters that govern the system with respect to time, relating to each other in a complicated manner.

The unsteady aerodynamics analysis can be done by solving the Navier-Stokes equation. However, the complexity of the equation makes it beyond current computational capabilities in most practical cases. To resolve this issue, one branch of study focused on dimensionality reduction and reduced order modeling. Traditional approaches in this branch include Proper Orthogonal Decomposition (POD)[1][2], Dynamic Mode Decomposition (DMD)[3], and Koopman analysis[4]. The problem with traditional approaches, which are mainly dealing with linear domains, is that model reduction in such a manner inevitably causes information loss. In the case of unsteady flows which require accurate prediction of transient phenomena, this is a severe issue.

Based on the universal approximation theorem, one can create neural network models that can approximate any given function. This led to the idea that a sufficiently deep neural network can approximate certain functions that can perform dimensionality reduction and reduced order modeling for complex flows with less information loss. In this context, there have been numerous studies recently, that explored the world of neural-network-based approaches for dimensionality reduction, reduced order modeling, and flow-related quantity prediction. Eivazi et al [5] performed dimensionality reduction using a neural network autoencoder. The autoencoder was fed with flattened 2D velocity magnitude and was trained to perform with average 93% reconstruction accuracy. Going one step further, he tested the performance of the autoencoder-LSTM framework for the prediction of the future state of the velocity magnitude for wake behind three cases, two types of fixed cylinders, and one oscillating airfoil. It was shown that the autoencoder-LSTM method outperformed the DMD method, and the performance gap was much higher for complex cases (cylinder case II and oscillating airfoil) with multi-frequency and extreme events phenomena. Maulik et al [6] focused on advection-dominated systems, which are challenging for the traditional method of POD & Galerkin projection due to their high dimensionality. They used a

convolutional autoencoder (CAE) for the reduced order modeling and rebuilt the system using LSTM, and showed that the proposed method is novel both in stability and convergence time. Viquerat et al [7] gave insight into a new approach for the prediction of drag for various shapes of airfoils. It created 9,600 random smooth airfoil shape images to train a model with convolutional layers plus fully connected layers, to be used to predict the drag of more realistic shapes. He achieved maximal relative error in the range of 1-2% for NACA airfoils.

Another strength of the machine-learning-based approach is in classification. Machine learning methods are superior in finding patterns that are difficult to figure out, due to their model-free nature. So, people explored the field of wake classification using ML, hoping to predict the upstream disturbance just by classifying the wake. Fukami et al [8] have experimented on four supervised machine learning techniques in wake classification, to give general guidelines in terms of accuracy, computational cost, and robustness for canonical flow problems. It has been found that ELM (Extreme Learning Machine) worked best for wake classification behind 2D cylinders considering accuracy and computational cost. However, the ELM was not robust enough when performing for wake behind 2D NACA0012 with a Gurney flap. Ribeiro et al [9] examined to give insight on wake similarity between various airfoil kinematics by ML-based clustering to be used in predictive models in energy harvesting. He used 2D vorticity magnitude of the wake behind the foil as input. For the classification model, he used 2D convolutional layers to extract the most important features of the vorticity. Then, by using LSTM (long short-term memory networks) he classified the input sequence of 5 samples into 4 classes. These 4 classes were determined via visual inspection. This paper has proven that properly divided classes can be classified via ML models achieving an average of 90% accuracy.

We focus on one type of neural network aforementioned in the references above, convolutional neural networks (CNNs). CNNs are promising when dealing with high input dimensions, as it is relatively free from many problems that can arise in deep neural networks with high input dimensions such as gradient vanishing. Due to this characteristic, CNNs have recently proven to be very powerful in image classification, starting from the project of classifying handwritten numbers[10]. The usage of CNNs for flow system modeling & analysis is beneficial as one can implement nonlinearity via nonlinear activation functions, to lower the errors. [11][6] CNNs are not only capable of dimensionality reduction and reduced order modeling, but also prediction of flow-related quantities.[12][13][7] This is especially powerful compared to a recurrent neural network (RNN) based approaches as CNNs can predict the desired quantity from instantaneous measurement.

The weakness of CNNs is that it is crucial to feed the network with vast, high-quality image data. This has been a big problem, as it is relatively difficult to create vast high-quality data of fluid. Nowadays, however, techniques to create such images have been sophisticatedly developed, and keep evolving. Starting from the development of Particle Image Velocimetry(PIV), many researchers in the field are striving for super-resolution imaging [14] and denoising algorithms [15], giving the possibility to overcome the weakness of CNNs.

To summarize, there is a promising possibility for CNNs to perform well in dimensionality reduction, reduced order

modeling, and flow-related quantity prediction for complex, high-dimensional flows with inherent nonlinearity. In this background, we focused on an inverse problem to estimate the sinusoidal airfoil pitching kinematics using CNN from the instantaneous velocity field measured in the wake far from the airfoil. In addition to simply being able to predict the airfoil kinematics from limited instantaneous velocity field measurements downstream of the airfoil, we aim to explore the effect of the number of sensing points by changing the measuring field area and downsampling the input image. Then, we studied the prediction of non-sinusoidal pitching cases, using a linearly increasing frequency pitching, or chirp signal, to explore the feasibility of the CNN method in more broad case.

II. Methodology

A. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a type of artificial neural network. Instead of updating the weights and biases, CNNs train to optimize *filters*. The basic framework of CNNs is shown in fig.1 In this basic framework, there is an original input image that is 3x3, 1 dimensional. In the convolutional layer, the filter size, which is 2x2, sets the range of input image to be read in each *convolutional calculation*. (*Convolutional calculation* outputs summation of element-wise multiplication.) Then each filter performs *convolutional calculation*, by moving through the original image at the size of *stride*. For example, filter 1 performs *convolutional calculation* four times, which creates $b1+d1$, $a1+c1$, $b1+d1$, and $a1+c1$, respectively. Then, the filtered image goes through the activation function, making the *feature map*. The feature map then goes through the batch normalization layer, which is proven to make the model to be faster and more stable [16], and the max pooling layer, which decreases the size of the feature map.

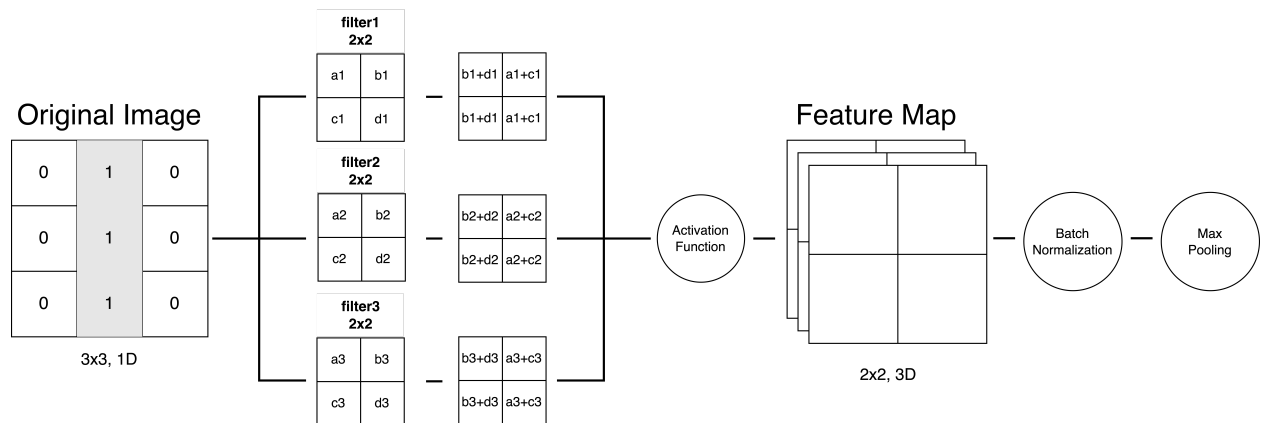


Fig. 1 The basic framework of the convolutional layer that has three 3x3 filters, followed by batch normalization layer and max pooling layer.

We used CNN architecture shown in fig.2. There are seven convolutional layers (conv2d layer) followed by one fully connected layer and the output layer.

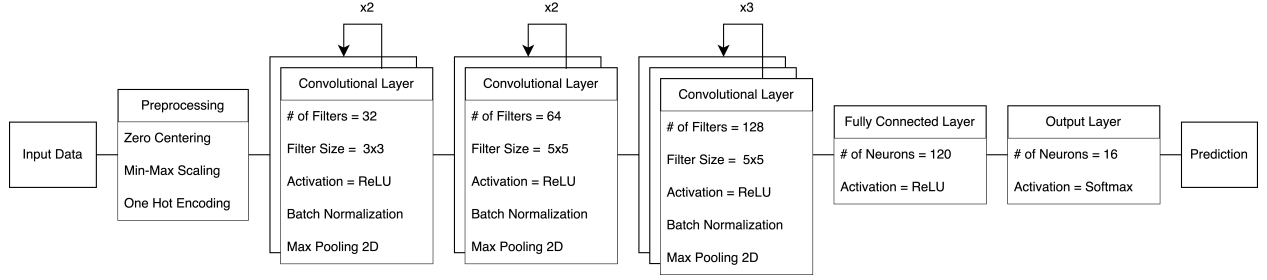


Fig. 2 The schematic diagram of CNN architecture used in this literature.

B. Classification with sparse basis selection

As a benchmark for comparison, we also use an alternative classification methodology, which is described here.

C. Direct Numerical Simulation

The database created by Towne et al., available at <https://deepblue.lib.umich.edu/data/collections/kk91fk98z?locale=en> was used to train and test CNN. [17](part V). Example snapshot of this database is shown in fig.3 The corresponding database is created by numerically solving the Navier-Stokes equation using immersed boundary projection method (Direct Numerical Simulation (DNS)). It models sinusoidal pitching airfoil in low Reynolds number flow ($Re = 100$), where Reynolds number was defined from freestream velocity (U_∞) and chord length (c). The motion of the airfoil is:

$$\alpha(t) = \alpha_0 - \alpha_p \sin(2\pi f_p t) \quad (1)$$

The database provides two base angles of attack, $\alpha_0 = 25^\circ$ and 30° , with the pitching amplitude $\alpha_p = 5^\circ$. For the frequency, the database provides 8 dimensionless frequencies $f_p = f c / U_\infty$ in set $\{0.05, 0.1, 0.2, 0.25, 0.3, 0.35, 0.4, 0.5\}$. The dimensionless timestep was defined as $0.01 c / U_\infty$, and the measurement was taken every 10 dimensionless timesteps. The stream-wise domain length is $12 \times c$, and wall-normal length of domain is $6 \times c$, making 600×300 grid points. The outputs used to train CNN were U_x (streamwise velocity) and U_y (wall-normal velocity). A more detailed explanation of the dataset can be found in [17].

For additional database for different cases, the same methodology was used. Later in the study, we expanded our idea to varying frequency pitching, or chirp signal pitching. We set everything except the frequency to be the same: $\alpha_0 = 25^\circ$ and 30° , $\alpha_p = 5^\circ$, dimensionless timestep = $0.01 c / U_\infty$, and so on. The dimensionless frequency was set to increase from 0.05 to 0.5 linearly over timestep. The equation for this kinematic of airfoil is shown in eq.2 3.

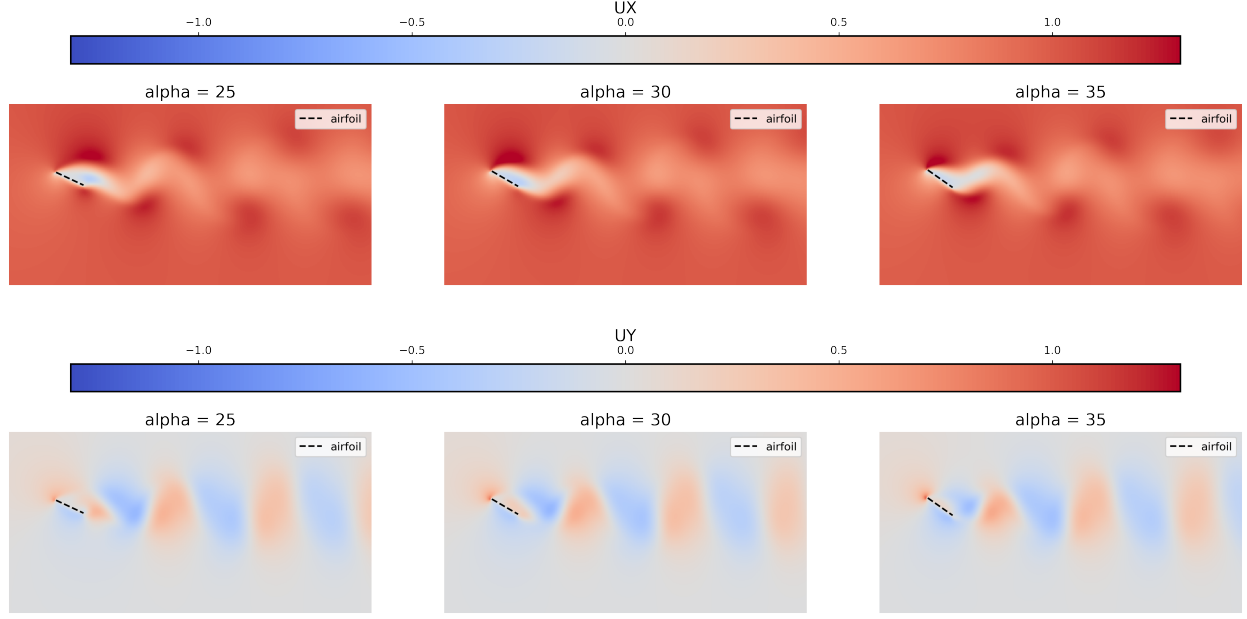


Fig. 3 The example of instantaneous dimensionless velocity field for UX and UY at 25°, 30°, and 35°. Base angle of attack = 30°, pitching amplitude = 5°, dimensionless pitching frequency = 0.25

t represents the dimensionless timestep, and t_{end} sets the timestep at which dimensionless frequency becomes the maximum, $f_1 = 0.5$. f_0 is the starting (minimum) dimensionless frequency, which is 0.05 in this case. We tested for $t_{end} = 6000$ and 9000, with measurement being done every 10 dimensionless timesteps.

$$\omega(t) = f_0 + \frac{t}{t_{end}}(f_1 - f_0) \quad (2)$$

$$\alpha(t) = \alpha_0 + \alpha_p \cos(2\pi\omega t) \quad (3)$$

D. Creating the Dataset and Preprocessing

In this literature, we first explored the effect of input image size on the prediction performance of CNN. For this testing, we used the sinusoidal pitching dataset created from DNS with two base angles of attack and eight dimensionless frequencies [17]. By doing this, we aim to see the feasibility of CNN on estimation of airfoil pitching kinematics, and furthermore novel way of reducing input image size. The second part of the study is where we expanded the idea of categorizing airfoil kinematics using CNN to varying frequency pitching motion prediction. For this part, we used a chirp signal pitching of airfoil, which was created by aforementioned.

The basic approach to making an input image for CNN is shown in fig. 4. A certain size of window was taken from the full data size of 599×299 , for example 200×100 as shown in black rectangle of 4. Note that the velocity field was taken from the wake farthest from the airfoil. Then, these two extracted window of measurement were concatenated

vertically, creating a square-shaped image of 200×200 . This method of concatenating UX and UY vertically to make a square image was applied throughout the study.

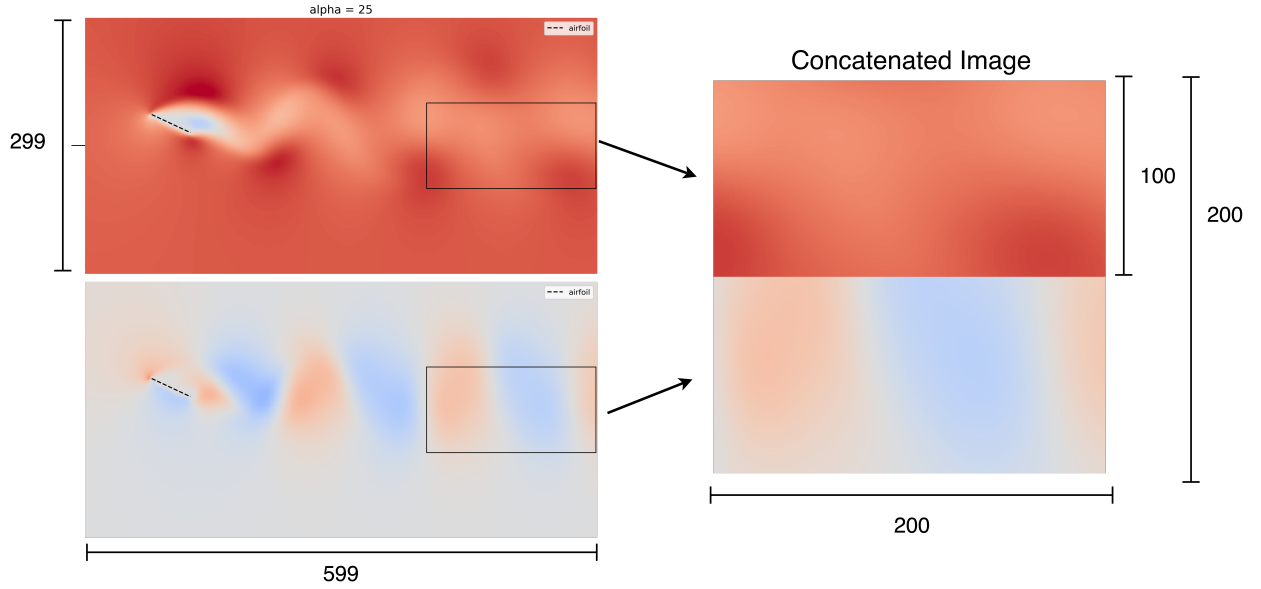


Fig. 4 The schematic illustration of the process to create a square-shaped input image.

1. Effect of Input Image Size - Sinusoidal Pitching

To test the effect of input image size, two types of image size reduction technique was used, denoted reduction 1 and reduction 2.

Reduction 1 is reducing measuring field area (or window size), meaning that we decreased the size of the black rectangle shown in fig. 4 on and on. For this literature, we will be quantifying this reduction in ratio, which corresponds to in what ratio the edge length was reduced. For example, if the ratio is $1/3$, this means that the full image of 600×600 was reduced into 200×200 . The tested ratios are $1/3$, $1/4$, $1/5$, $1/6$, $1/7$, $1/8$, $1/9$, $1/10$, $1/11$, $1/12$, $1/13$, $1/14$, and $1/15$, which are square images of edge length 200, 150, 120, 100, 86, 76, 66, 60, 54, 50, 46, 42, and 40, respectively. Note that the number of sensing points was reduced in squared values of ratios.

Reduction 2 is simplified image downsampling. The schematic of this process is shown in Fig. 5. The downsampling was done by skipping a number of pixels before collecting the next one. This number of pixels skipped is *strides*. The images from reduction 1 were used as the original image before downsampling. Here, we denote the size of the image before downsampling as the base window ratio. For example, if the base window ratio was $1/3$ and stride = 1, then the input image size becomes that equivalent to window ratio of $1/6$, making 100×100 . If the base window ratio was $1/4$ and stride = 1, it will be 76×76 . The base window ratios of $1/3$, $1/4$, $1/5$, $1/6$, and $1/7$ was used, and all the tested cases are listed in table 1 for better understanding. Note that due to the CNN architecture explained above, the minimum edge length tested was 40, as smaller images cause negative dimension size.

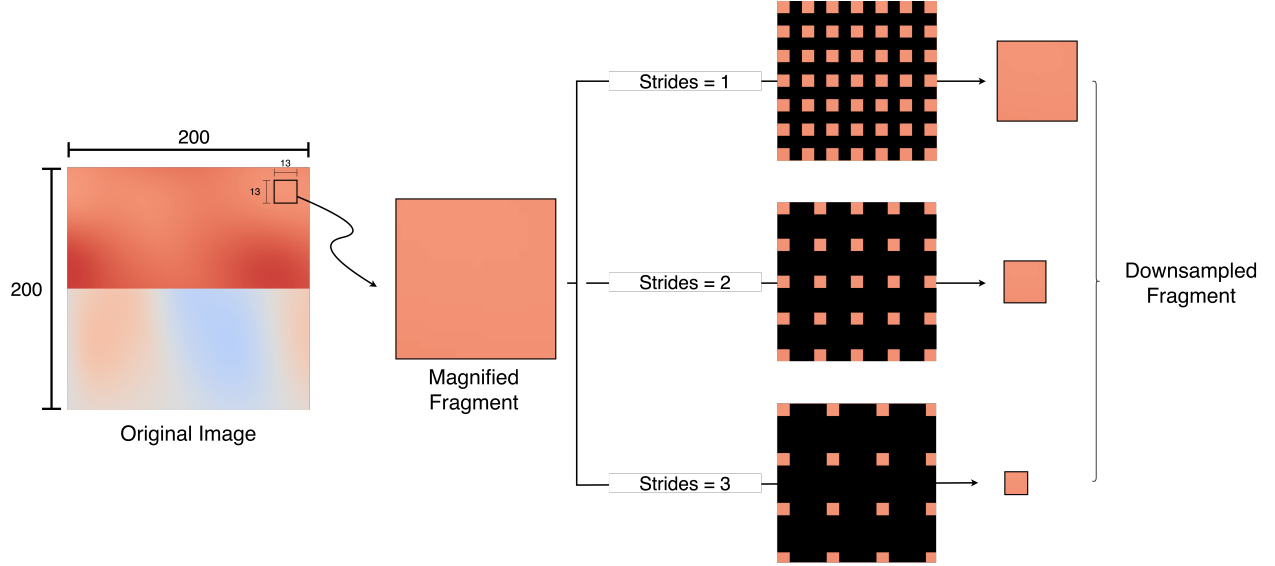


Fig. 5 The schematic of the process to create downsampled images according to strides value. The example is shown for a fragment of the whole image for better understanding.

Table 1 Tested cases for test 2

Base Window Ratio	Strides	Edge Length
1/3	1	100
1/3	2	66
1/3	3	50
1/3	4	40
1/4	1	76
1/4	2	50
1/5	1	60
1/5	2	40
1/6	1	54
1/7	1	42

For both reduction in sinusoidal pitching, all the datasets were split into train, validation, and test data with stratification with base angle of attack and frequency. The ratio was train : validation : test = 70 : 15 : 15, which makes 7856, 1680, and 1680 images, respectively. Three types of processing were done: zero-centering, scaling, and weighting. For zero-centering, we obtained mean image from training data, and subtracted the mean image from training, validation, and test data. The scaling was needed to be done since the velocity range for the UX and UY was very different, as in fig. 3. Therefore, the UX and UY velocity value were scaled using minimum and maximum values of them, as in ???. The minimum and maximum values were calculated from training data, and these values were used to scale all training, validating, and testing data. Lastly, weighting was done as there were more data from 30° than in 25° cases for the sinusoidal pitching cases. When training the neural network model, the loss from 25° cases was weighted by the factor of 2.5 accordingly.

$$UX = UX / (UX_{max} - UX_{min})$$

$$UY = UY / (UY_{max} - UY_{min})$$

2. Expansion to Varying Frequency Pitching - Chirp Signal Pitching

The image was created by concatenating U_X and U_Y vertically, as for the sinusoidal cases shown in fig. 4. However, for chirp signal pitching, we only used one input image size, which was measuring field area of 200×200 with no stride. The chirp signal created has a continuous, varying frequency. However, we need a regression model to work with continuous frequency, which is very different from classification models. So, to utilize the original classification model, the continuous frequency of the chirp signal was categorized into $\{0.05, 0.1, 0.2, 0.25, 0.3, 0.35, 0.4, 0.5\}$. This was done by taking the average values in two consecutive categories and using them as the boundary values. For example, frequency in range $0.05 \leq \omega < \frac{0.05+0.1}{2}$ was categorized into label 0.05, frequency in range $\frac{0.05+0.1}{2} \leq \omega < \frac{0.1+0.2}{2}$ was categorized into label 0.1, etc. The range for categorization is shown in fig. 6.

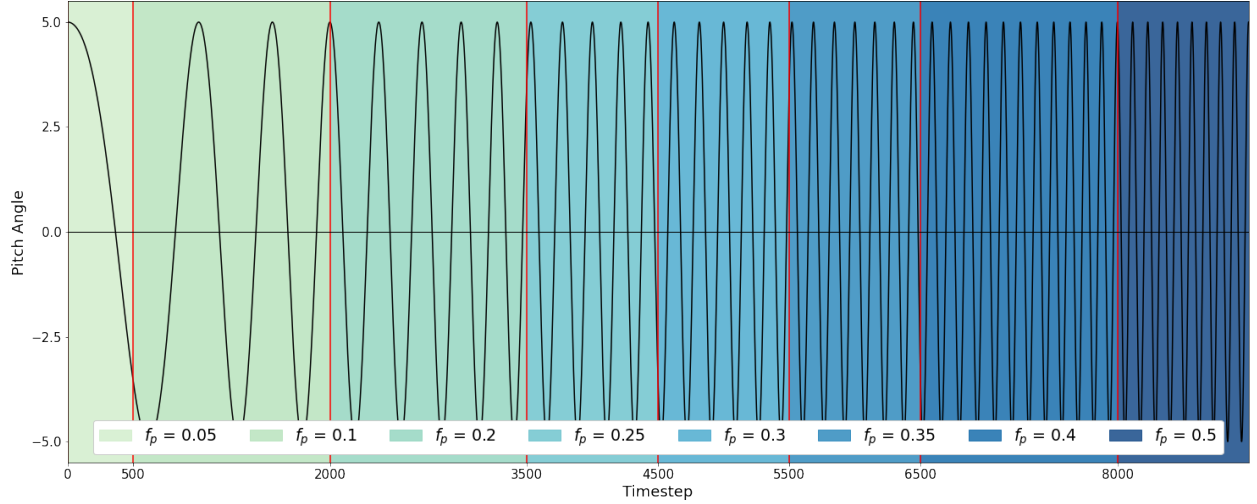


Fig. 6 The range of dimensionless frequency for the categorization of chirp signal.

The train-test split was done in different ways as there were three types of test done for the chirp signal pitching: Test 1. Evaluation of Chirp Signal on Model Trained from Sinusoidal Pitching Data, Test 2. Evaluation of Chirp Signal on Model Trained from Sinusoidal + Chirp Data, and Test 3. Evaluation of Out-of-Distribution Chirp Signal on Model Trained from Sinusoidal + Chirp Data. For test 1, the model trained from sinusoidal pitching data of reduction 1 was used (the ratio was 1/3). So, all the data generated for the chirp signal of $t_{end} = 9000$ was used for testing, making training : testing = 7856 : 1798. In test 2, a new model was trained using both sinusoidal data and chirp signal data. So the data split was done by train : validation : test = 70 : 15 : 15, making 9109 : 1953 : 1952. Note that in test 2, the

ratio of sinusoidal : non-sinusoidal was about 6:1. This will be an imbalanced dataset, and this is intended as test 2 was done to make the model learn the sense of varying frequency while being able to classify sinusoidal pitching well. In test 3, the dataset for training and testing was different from in test 1. The training data was created in test 2, which was 9109 data, and all the data generated for chirp signal of $t_{end} = 6000$ was used, which was 1198 data. Zero-centering and scaling were done in the same manner as in sinusoidal pitching, defining the mean image and scaler from training data and applying it to the validation and test data.

E. Testing the Performance of CNNs

For every tested case, the performance of the model was tested mainly on accuracy. Furthermore, the confusion matrix for each case was examined to avoid any biased results and get insight into which categories are most confusing for the model. Due to the random nature of the training neural networks, repeating the training produces a slightly different model. So, we tested for one neural network training cycle, the case from test 1 of ratio 1/9. The results are shown in table 2. We found that the accuracy remains in the range of 94.52-98.33% when repeating the training process 10 times with different random seeds. The average accuracy was 96.60%, and the standard deviation was 1.09%. The range of accuracy was not subtle enough to ignore the effect of randomness, so all the evaluation of performance was done on 10 CNNs trained in different random seeds, and we used the averaged results for every case.

Table 2 Results for one neural network training cycle - test 1, ratio: 1/9

Trial #	Accuracy(%)	Deviation(%)	Trial #	Accuracy(%)	Deviation(%)
1	95.24	-1.36	6	96.37	-0.23
2	97.20	+0.6	7	97.38	+0.78
3	94.52	-2.08	8	95.95	-0.65
4	96.79	0.19	9	98.33	+1.73
5	96.55	-0.05	10	97.68	+1.08

III. Result

A. Sinusoidal Pitching

1. Reduction 1. Reducing Window Size

The result for reduction 1 is shown in fig.7. The x axis represents the ratio of reduction, which are 1/3, 1/4, 1/5, 1/6, 1/7, 1/8, 1/9, 1/10, 1/11, 1/12, 1/13, 1/14, and 1/15. They correspond to square images of edge lengths 200, 150, 120, 100, 86, 76, 66, 60, 54, 50, 46, 42, and 40, respectively. Note that the total number of measurement points is ratio of reduction squared. The y-axis represent accuracy, ranging from 85% to 100%. The black dots on each ratio represent all ten accuracy emerging from the random nature of the neural network, and the blue plot is the averaged accuracy for the black dots in each ratio. Curve-fitting was done for the averaged accuracy, and it is shown in the red curve.

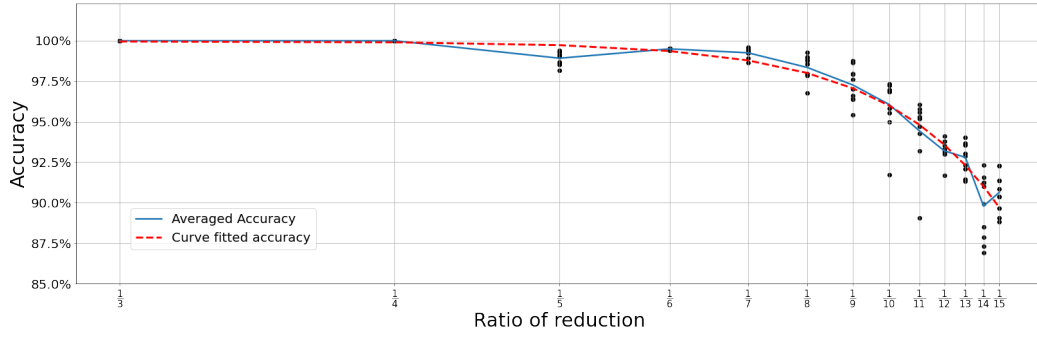


Fig. 7 The result for the reduction 1.

The accuracy in reduction 1 dropped exponentially as ratio decreased, following the curve fitted equation 4. The coefficient of determination (R^2), was 0.9752

$$accuracy = -0.6846e^{-28.48 \times \text{ratio of reduction}} + 0.9996 \quad (4)$$

Although the accuracy drops exponentially, the averaged accuracy remained in approximately over 90%. Prediction of airfoil kinematics could be done with high accuracy even in substantial reduction of window size. We could also see that the accuracy range of difference, shown in black dots, increasing as the ratio became smaller. This seems to be natural, since the model is being provided with less data inputs as the input image size decreases. This can make the stochastic process of the model to affect the performance more heavily across different training runs. Note that this trend of standard deviation increase should only be interpreted in cases of relatively lower accuracy, or ratios under 1/8, because when the prediction is too easy for the model, there is no room for the accuracy to have a great deviation. The low standard deviation in high accuracy cases are not being caused by this trend. The average accuracy and its standard deviation is shown in table 3.

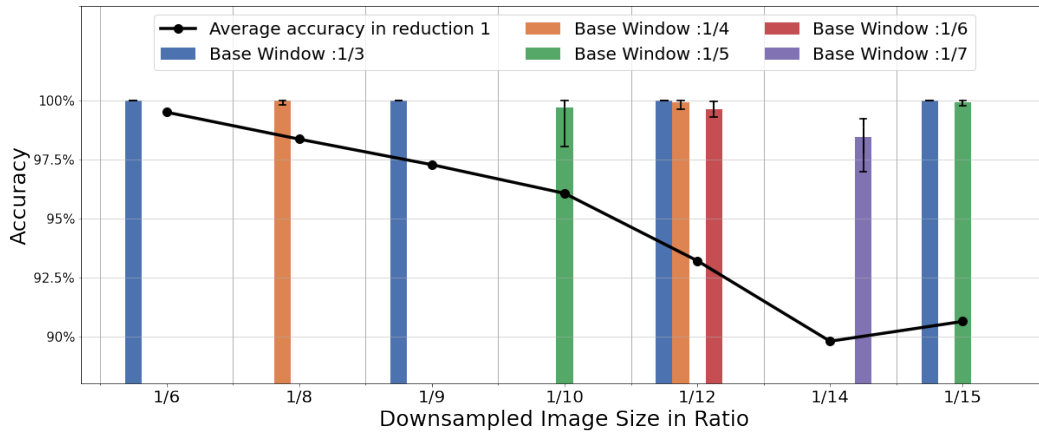
2. Reduction 2. Downsampling

The result of reduction 2 is shown in fig 8. The x-axis represents the final size of the image. This means that a base window ratio of 1/3 with strides = 1 will be in category 1/6 label, a base window of 1/4 with strides = 2 will be in category 1/12 label, and so on. The colors of the bar graphs represent the base window ratio before downsampling, and the y-axis shows the accuracy. The black plot with an 'o' shaped marker is the result from reduction 1, inserted for the comparison between reduction 1 and reduction 2. The black I-shaped line on each bar represents the error range of each bar, on 10 different training runs. The height of the bars shows the averaged accuracy. The values of the plot are listed in table 4.

The performance of airfoil kinematics prediction from reduction 2, downsampling, was extremely high. The

Table 3 The averaged accuracy and its standard deviation for all ratios of reduction

Ratio #	Averaged Accuracy(%)	Standard Deviation(%)
1/3	100	0
1/4	100	0
1/5	98.923	0.3889
1/6	99.500	0.0476
1/7	99.256	0.2705
1/8	98.357	0.6948
1/9	97.274	1.0264
1/10	96.060	1.6359
1/11	94.440	1.9587
1/12	93.202	0.6104
1/13	92.798	0.9241
1/14	89.792	1.8725
1/15	90.631	1.1576

**Fig. 8 The result for the reduction 2.**

lowest case was averaged accuracy of 98.458%, base ratio 1/7, strides 1. The model performance was nearly perfect, approaching nearly 100% for all tested cases.

Especially when focusing on the image size 1/12 column, which consists of three bars, we could find that the performance of base window = 1/3 was higher than base window = 1/4, and the performance of base window = 1/4 was higher than that of base window 1/6. Having a wider area of measurement, or base window ratio, governed the performance of CNN. This could also be seen from the fact that having a larger base ratio performed better than that of a smaller base ratio, regardless of the final image size. For example, base ratio = 1/3 (blue bar) always performed better than that of base ratio 1/4, 1/5, 1/6, and 1/7, the averaged accuracy of base ratio = 1/4 (orange bar) always performed better than that of 1/5, 1/6, and 1/7, the averaged accuracy of base ratio = 1/5 (green bar) always perform better than that

Table 4 The averaged accuracy and its standard deviation for all ratios of reduction

Final Image Size	Base Ratio	Strides	Averaged Accuracy (%)	Standard Deviation (%)
1/6	1/3	1	100	0
1/8	1/4	1	99.982	0.054
1/9	1/3	2	100	0
1/10	1/5	1	99.708	0.58
1/12	1/3	3	100	0
1/12	1/4	2	99.929	0.143
1/12	1/6	1	99.637	0.193
1/14	1/7	1	98.458	0.649
1/15	1/3	4	100	0
1/15	1/5	2	99.923	0.088

of smaller base ratio, and so on. While the base window size dominated, the downsampling of the image had very little effect on the accuracy of the model. Looking closely at the height of the bars in same base ratio, the difference in accuracy from downsampling was extremely small, having a maximum difference of 0.215% for base ratio 1/5 with strides 1 and 2.

Note that the standard deviation was very small in downsampling reduction as well, compared to that of reduction 1. However, this seems to be simply coming from the fact that there is no room for variation to be large when the accuracy is always near 100%. This should not be taken into interpretation more than mere verification of the result.

B. Classification with Sparse Basis Selection

We also tested an alternative classification methodology as a benchmark, using sparse basis selection. The result of this method in comparison with CNN model in reduction 1 is shown in table 5. We can note here that the CNN models, are performing with much higher accuracy. Especially when decreasing the measuring area, or the window size, the sparse basis selection model are influenced significantly more than the CNN models. This shows that sparse basis selection method will not be feasible for applications that can only get measurement from a small window, or applications that require significantly high accuracy.

Table 5 The prediction accuracy comparison between CNN model and sparse basis selection method

Ratio	CNN Model Accuracy (%)	Sparse Basis Selection Method Accuracy (%)
1/3	100	88.571
1/6	99.500	60.714
1/9	97.274	45.060
1/12	93.202	26.726
1/15	90.631	28.869

Also, there is a trade-off between training time and predicting time. For CNN models, it takes significant time to

train the model; while the prediction time is relatively short. The sparse basis selection method has opposite in time consumption. It takes relatively short time to create library for all the basis, but it takes much longer time to predict in which category the input image fall in. This is largely due to the nature of sparse basis selection prediction method, which is solving an optimization problem. This could be detrimental to a system which require a fast feedback for control compared to the CNN model.

However, the sparse basis selection model has its strength in predicting with only few given measurement. While CNN models can perform with inputs that are significantly smaller measuring field area compared to sparse basis selection, the model is infeasible if there are only few measurements, e.g. 50 measuring points. In [17], the prediction of airfoil pitching kinematics was done using only 50 measuring points that have been randomly selected, scoring overall accuracy of 78%. Although the accuracy is lower compared to the CNN model, considering the significant amount of input measuring points, sparse basis selection model clearly has its strength and suitable applications.

C. Chirp Signal Pitching

1. Test 1 - Evaluation of Chirp Signal on Model Trained from Sinusoidal Pitching Data

The prediction of chirp signal with $t_{end} = 9000$ was tested on a model trained from sinusoidal pitching of ratio 1/3. The average accuracy was 27.4% with a standard deviation of 0.397%. The varying frequency data was out-of-distribution data for sinusoidal, which we anticipated for the CNN to be able to find the underlying pattern in the same base angle of attack and frequency. However, the accuracy was extremely low. The confusion matrix 9 also shows that the model was not able to estimate the kinematics, having no trend of prediction in diagonal entries. This was tested for a more strictly categorized frequency as the low accuracy may be coming from the large variance of continuous frequency values being categorized into the same category. We allowed only 0.01 dimensionless frequency difference between categorized frequency and continuous frequency, and the average accuracy was 17.9 %. The confusion matrix for this strictly categorized case is here: 10. The accuracy got even lower, and the confusion matrix still doesn't show a clear trend in the vicinity of diagonal entries. So we confirmed that categorization was not the problem.

2. Test 2 - Evaluation of Chirp Signal on Model Trained from Sinusoidal + Chirp Data

Seeing that the prediction of the chirp signal couldn't be done on models trained on sinusoidal data, a new model was trained using sinusoidal data and chirp signal of $t_{end} = 9000$ together. The ratio of sinusoidal : chirp was about 6:1, making it highly imbalanced data between sinusoidal and non-sinusoidal data. This was tested as mixing a small amount of non-sinusoidal data in training was expected to give robustness toward frequency variation while maintaining the performance of sinusoidal pitching prediction. The accuracy for sinusoidal pitching case remained in average accuracy of 99.8% with a standard deviation of 0.168%. For the chirp signal, the average accuracy was 95.5% with a standard deviation of 2.93%. This was a great increase in accuracy for the chirp signal, and the accuracy of sinusoidal pitching

prediction remained near perfect. Here are two confusion matrices for the prediction of non-sinusoidal pitching and sinusoidal pitching. 11 12 Test 2 proved us the feasibility of CNN to predict the varying frequency as well when it was fed the data.

3. Test 3 - Evaluation of Out-of-Distribution Chirp Signal on Model Trained from Sinusoidal + Chirp Data

As the prediction of the chirp signal on the model trained from sinusoidal + chirp data was performing well, we took a step further to see the performance of the chirp signal with $t_{end} = 6000$ (frequency changing faster). We were anticipating that training with chirp signal with $t_{end} = 9000$ would implement the effect of increasing frequency, making the model to be able to characterize the kinematics of faster-increasing chirp signal ($t_{end} = 6000$). The average accuracy in test 3 was 29.4% and the standard deviation was 2.72%. This was frustrating to see that the model, even though implemented with varying frequency, couldn't predict the airfoil kinematics when the rate of change was slightly different. However, checking the confusion matrix 13, the prediction had a clear trend of staying near the diagonal line. This gave the possibility for prediction of chirp signal of different rates of change when implemented with the varying frequency in training if we could build a CNN model that was more robust.

This was tested for more strictly categorized frequency as well, to check if the error from categorizing the continuous frequency is causing the low accuracy. Allowing 0.01 frequency difference between categorized frequency and continuous frequency, the average accuracy for this case was 25.9% with a standard deviation of 5.08%. The error from categorization was not the reason. Here is the confusion matrix for strictly categorized cases: 14.

IV. Conclusion

In former studies, CNNs have been showing great performance in fluid mechanics for the analysis and prediction of complex, high-dimensional flows with inherent non-linearity. So in this article, we explored the estimation of airfoil pitching kinematics using convolutional neural networks. The estimation was done from the instantaneous velocity field snapshot downstream of the wake. The pitching kinematics was split into two types: sinusoidal pitching and chirp signal pitching.

For the sinusoidal pitching, we tested on two base angles of attack (α_0) of 25° and 30°, with pitching amplitude of 5° and dimensionless pitching frequency (f_p) of 0.05, 0.1, 0.2, 0.25, 0.3, 0.35, 0.4, and 0.5. The estimation was done by classifying the kinematics into the combination of α_0 and f_p , making it a total of 16 categories. The CNN was challenged by reducing the input image size in two ways: 1. reducing the measuring field area, and 2. downsampling the input image. In both reductions, the airfoil pitching kinematics could be estimated with high accuracy even with substantial reduction.

In reduction 1, the accuracy dropped exponentially as the input image size got smaller. However, even in the most extreme cases, accuracy remained approximately over 90% (the lowest case was 89.792%). The exponential drop

followed the equation 4. The results are plotted in 7, and listed in table 3. The performance of CNN and sparse basis selection method was compared, to show that the CNN models are a novel approach. As shown in table 5, the accuracy of CNN was significantly higher than that of sparse basis selection. Also, the adversarial effect of reduction of window size impacted sparse basis selection model much more significantly, showing that CNN models will be more suitable for applications that involve smaller measuring field area. Also, the prediction time for sparse basis selection method was too long compared to the training time. Considering that model deployment favors longer training - shorter prediction time, CNN methods also have an advantage in such sense. However, the sparse basis selection can work with only few measuring point data given while the CNN models need much more input data point, sparse basis selection may have advantages in situations involving only a few sensing points in a wider measuring field area.

Reduction by downsampling had a more fascinating result, having an accuracy of 98.458% in the lowest case. The model could estimate the base angle of attack and pitching frequency with very high accuracy. When comparing the same input image size with reduction 1, the accuracy was significantly higher when the reduction was done by downsampling. We found that the base window size was dominant for the performance of CNNs. Having a larger base window size with bigger strides performed better even if the final input image size was smaller when compared with a smaller base window size with smaller strides having a larger final input image size. This is well shown in 8, where the height of the blue bar (base ratio = 1/3) is always higher than that of the orange bar (base ratio = 1/4), the height of the orange bar is always higher than green bar (base ratio = 1/5), and so on. The difference in accuracy caused by downsampling for the same base window size was almost negligible compared to the effect of base window size. Examining the height of bars in same colors, the maximum difference in accuracy was 0.215% for base ratio 1/5 with strides 1 and 2.

Looking at results from the two types of reduction in sinusoidal pitching, downsampling (reduction 2) seemed to be worth more examination to test its limit in size of strides to maintain favorable accuracy. However, reducing the input image size has its limit when applied to CNNs. To reduce the input image size beyond a certain level, the depth of CNN architecture must be reduced as well, as CNNs work by decreasing the dimensionality of the input image layer by layer. Using the same deep CNNs in smaller input image size will result in a negative dimension, and thus is not feasible. This raises the question of the tradeoff between downsampling, or sparse sensing and model depth. Assuming that the measuring field area is sufficient, sparse sensing from the field will have a benefit in the cost of prediction. How far can we sparsely sample the velocity along with the reduction of CNNs's depth while maintaining a reasonable range of accuracy? This is the next step for the idea to possibly expand. Also, even the reduction of window size (reduction 1), although seemed less significant compared to downsampling, was performing nearly over 90%, meaning that the model needs to be challenged more in the same sense.

For the chirp signal pitching, we increased the pitching frequency linearly. There were two different rates of change for the frequency: $t_{end} = 9000$ and $t_{end} = 6000$. $t_{end} = 9000$ was relatively slowly increasing frequency and

$t_{end} = 6000$ had relatively faster increase. The pitching motion is described in 2.3. We tried three tests on the estimation of chirp signal pitching motion. The first test evaluated the estimation of chirp signal kinematics of $t_{end} = 9000$ on a model that was trained on sinusoidal pitching. The window size was 200×200 , which corresponds to the 1/3 ratio for the reduction of measuring field area. This was tested as we were anticipating that the model could perform proper wake classification from the fundamental difference in the base angle of attack and instantaneous frequency although the varying frequency can create big changes in the wake. However, the test result was an average accuracy of 27.4%, showing no remarkable performance in the estimation of pitching kinematics. The varying frequency seemed to be a clear out-of-distribution data for the model, as there was no clear trend of prediction being near the diagonal line in confusion matrix 9. Even when the continuous frequency was more strictly classified, the model was not able to estimate the kinematics, confirming that the classification of frequency was not the reason for the low performance.

The model trained on the sinusoidal data could not spot the fundamental similarity between sinusoidal pitching and chirp signal pitching. So, we moved on to the second test to check if the model could estimate the airfoil pitching kinematics when the model had a chirp signal in its training data. We used a mixture of sinusoidal pitching and chirp signal pitching with $t_{end} = 9000$, which was in a ratio of 6:1. This imbalance was intended to maintain the high performance on sinusoidal pitching while informing the model with a glimpse of chirp signal cases. This attempt was successful, recording an average accuracy for chirp signal pitching of 95.5%, and an average accuracy for sinusoidal pitching of 99.8%. The two confusion matrices are given here: 11 12. The result is very favorable as the accuracy for sinusoidal pitching remained in the region near perfect, while the chirp signal prediction improved significantly.

Having known that the model could learn the effect of varying frequency, we tested on a more rapidly changing chirp-signal-pitching, with $t_{end} = 6000$ (1.5 times more quickly increasing compared to $t_{end} = 9000$). The model could not perform well anymore, and the average accuracy was 29.4% in this case. From the confusion matrix 13, however, we could still see that the prediction stayed near the diagonal line of correct prediction. The model learned the chirp-signal pitching pattern in some way but lacked in generalizing the patterns. To check if the error from the categorization of continuous frequency is causing this error, we also tried for strict categorized data, with frequency error under 0.01, and the averaged accuracy was 25.9%. This proved that the categorization was not the problem. Here is the confusion matrix: 14.

Combining the result from the sinusoidal pitching and non-sinusoidal pitching, CNN models could perform nearly perfectly when provided the exact case in the training process. In CNN models that could perform perfectly, the input image size could even be significantly reduced, and still performed with high accuracy. This is promising as the basic concept of using an instantaneous snapshot of the velocity field in the wake far from the airfoil for the estimation of the pitching kinematics is feasible. However, the biggest problem with the CNN architecture that we used was the lack of robustness. This is not good for practical purposes, where we can frequently encounter out-of-distribution data. Also, the reduction of input image size in sinusoidal pitching inspired us to test the reduction of image size along with the

reduction of network depth, where shallower models are expected to have more robustness in prediction. This motivates us to the next step of estimating airfoil kinematics from a snapshot of the velocity field, to create a more robust model, with less depth, or possibly completely different ways.

Appendix

An Appendix, if needed, appears **before** research funding information and other acknowledgments.

Funding Sources

Sponsorship information and acknowledgments of financial support should be included here. **Authors are responsible for accurately reporting funding data relevant to their research.** Please confirm that you have correctly entered **all sources** of funding and grant/award numbers **for all authors** in this section of your article. You will also be asked to select the appropriate funding organization from a drop-down menu in ScholarOne when you submit your manuscript. Be careful to choose the correct funder name, as organization names can be similar, and also be mindful to select sub-organizations within the registry hierarchy that are the actual funding sources, as appropriate, rather than choosing the name of the parent organization. Information provided in your manuscript must match the funding data entered in ScholarOne.

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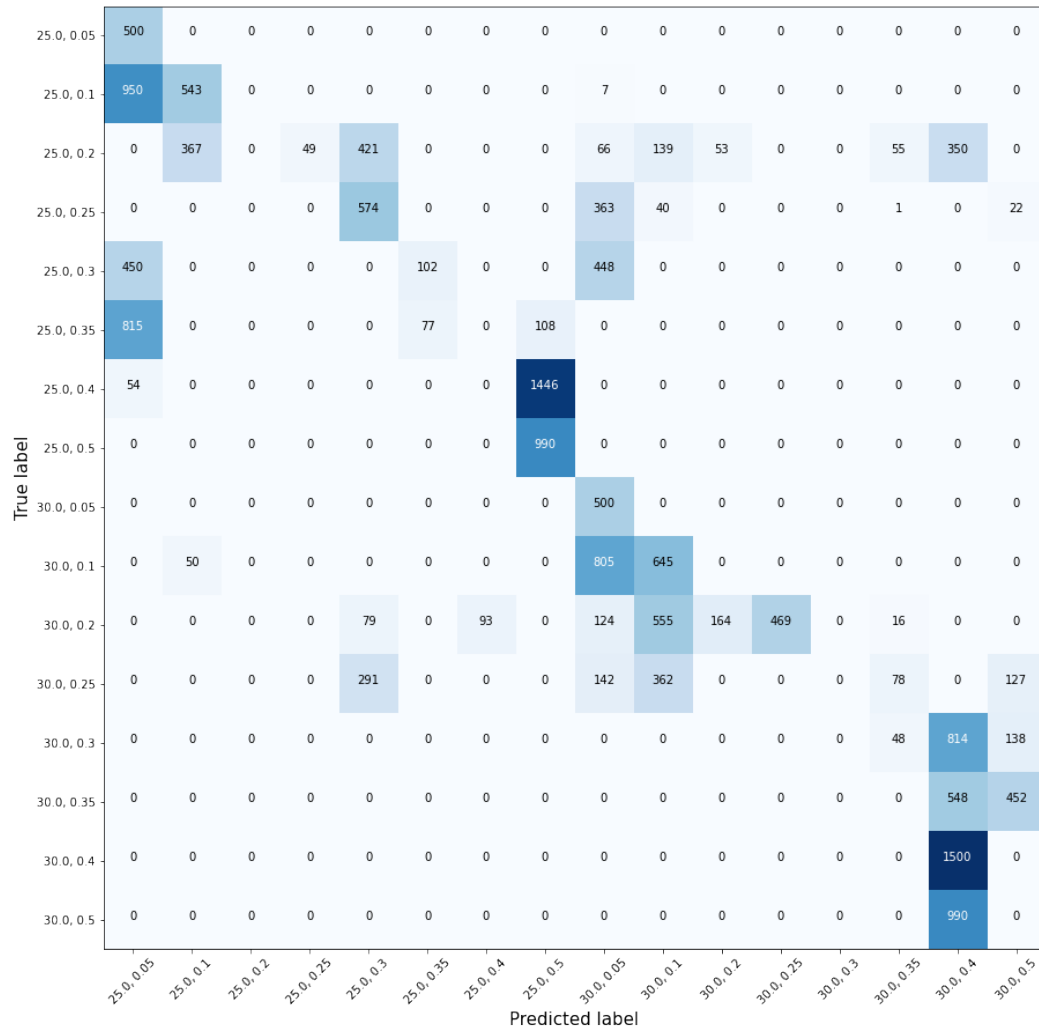


Fig. 9 Confusion matrix for the prediction of chirp signal on a model trained from sinusoidal pitching data

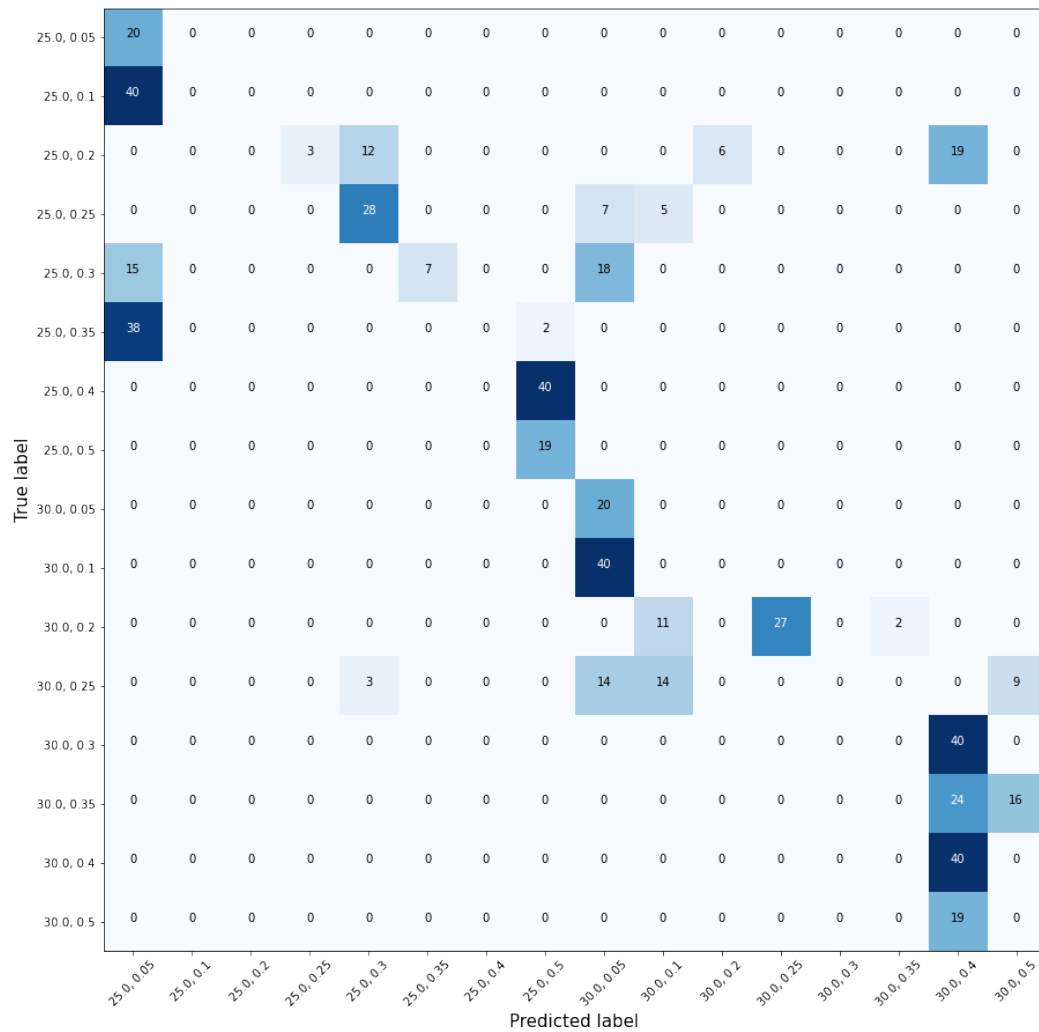


Fig. 10 Confusion matrix for the prediction of chirp signal on a model trained from sinusoidal pitching data, with stricter categorization of continuous frequency

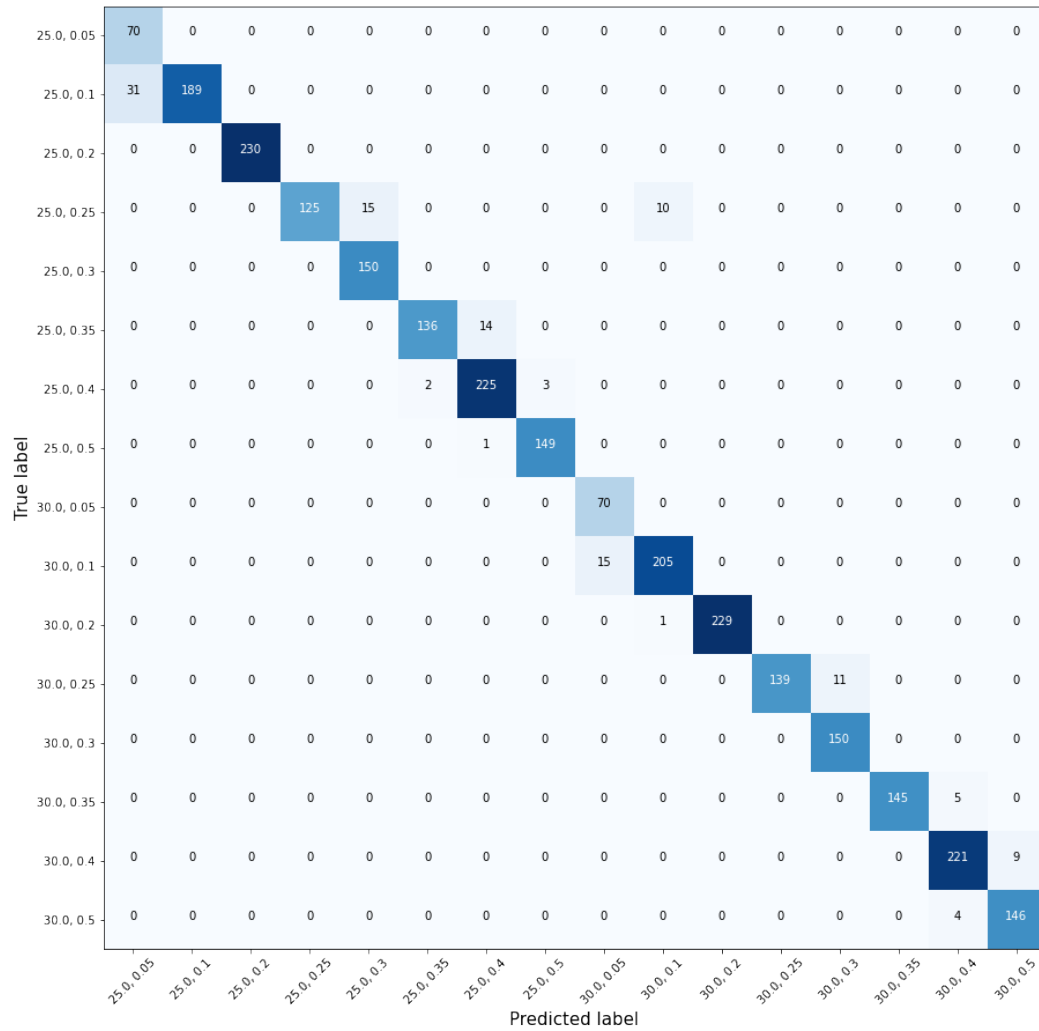


Fig. 11 Confusion matrix for the prediction of chirp signal on model trained from sinusoidal + chirp data

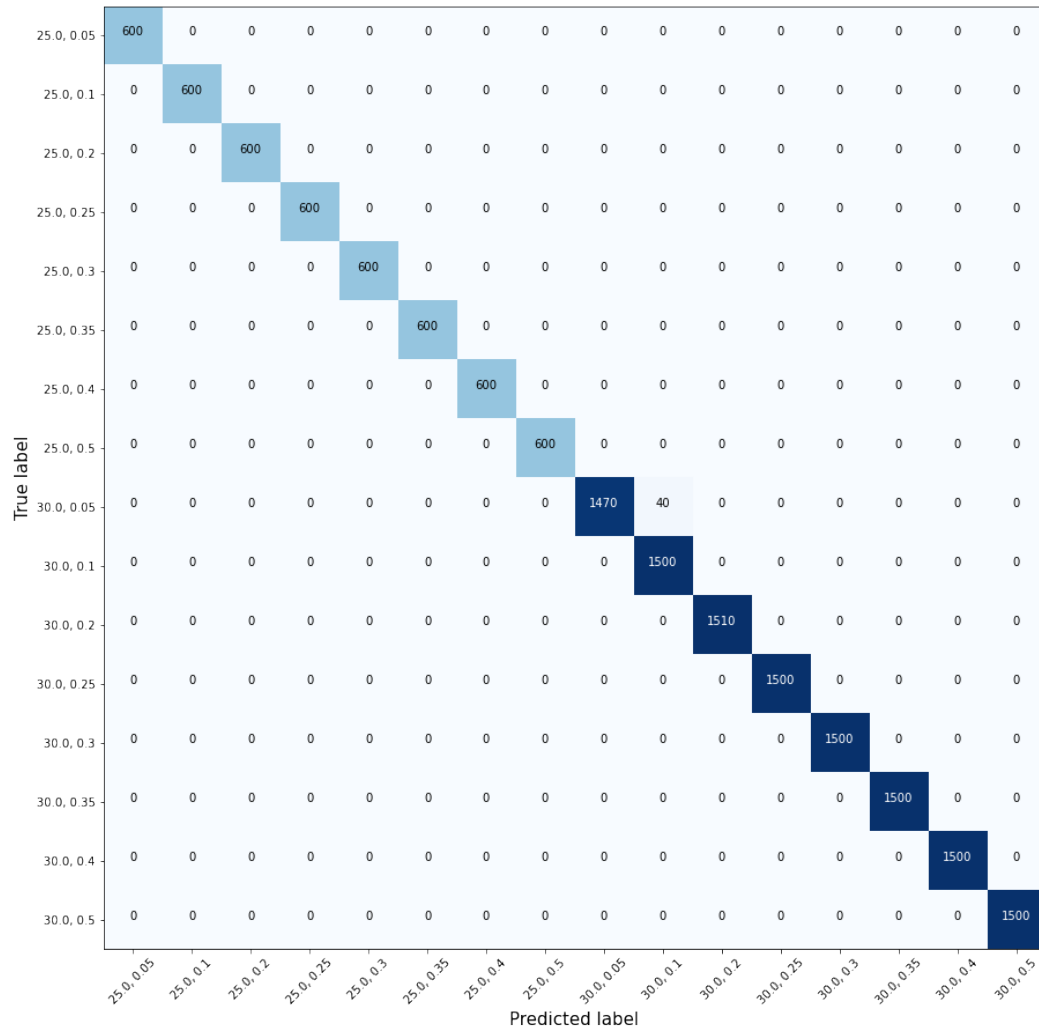


Fig. 12 Confusion matrix for the prediction of sinusoidal signal on a model trained from sinusoidal + chirp data

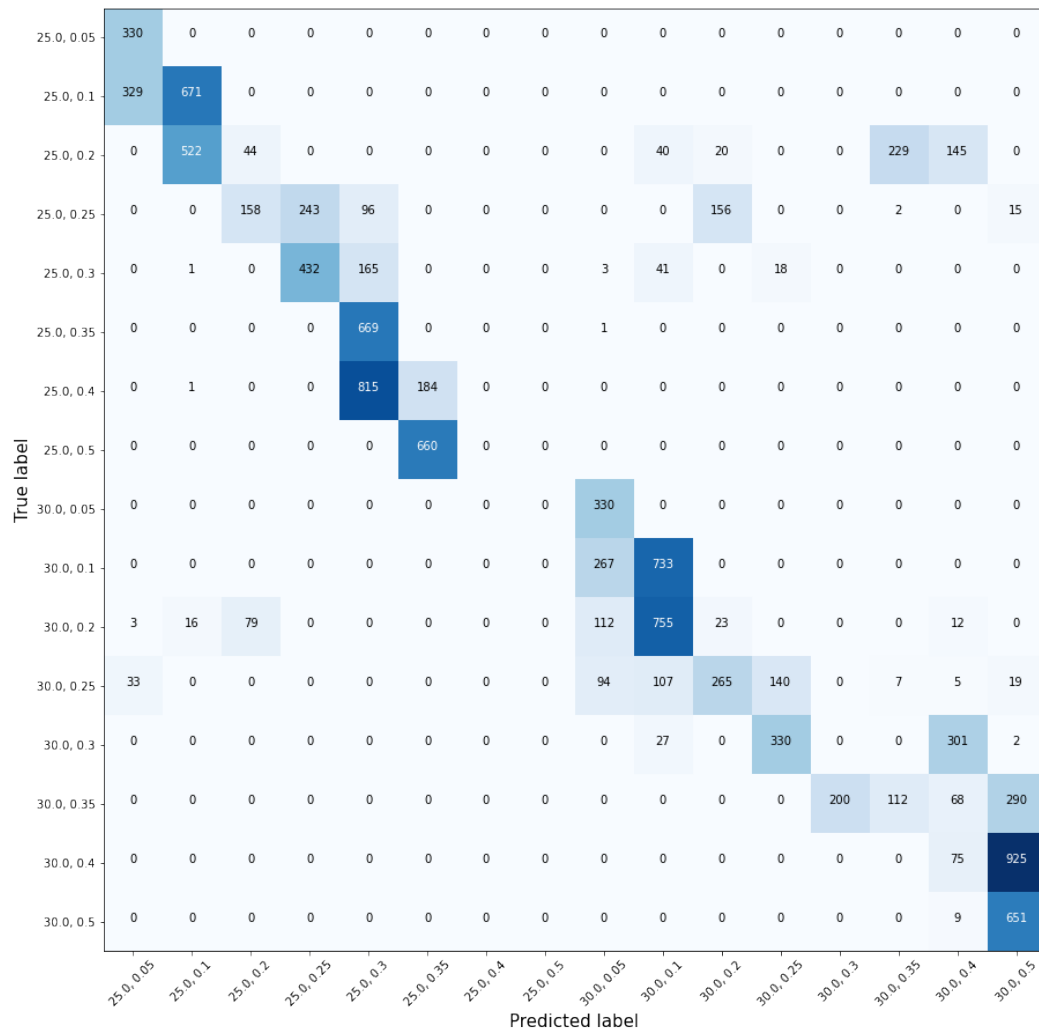


Fig. 13 Confusion matrix for the prediction of faster-changing chirp signal on a model trained from sinusoidal + chirp data

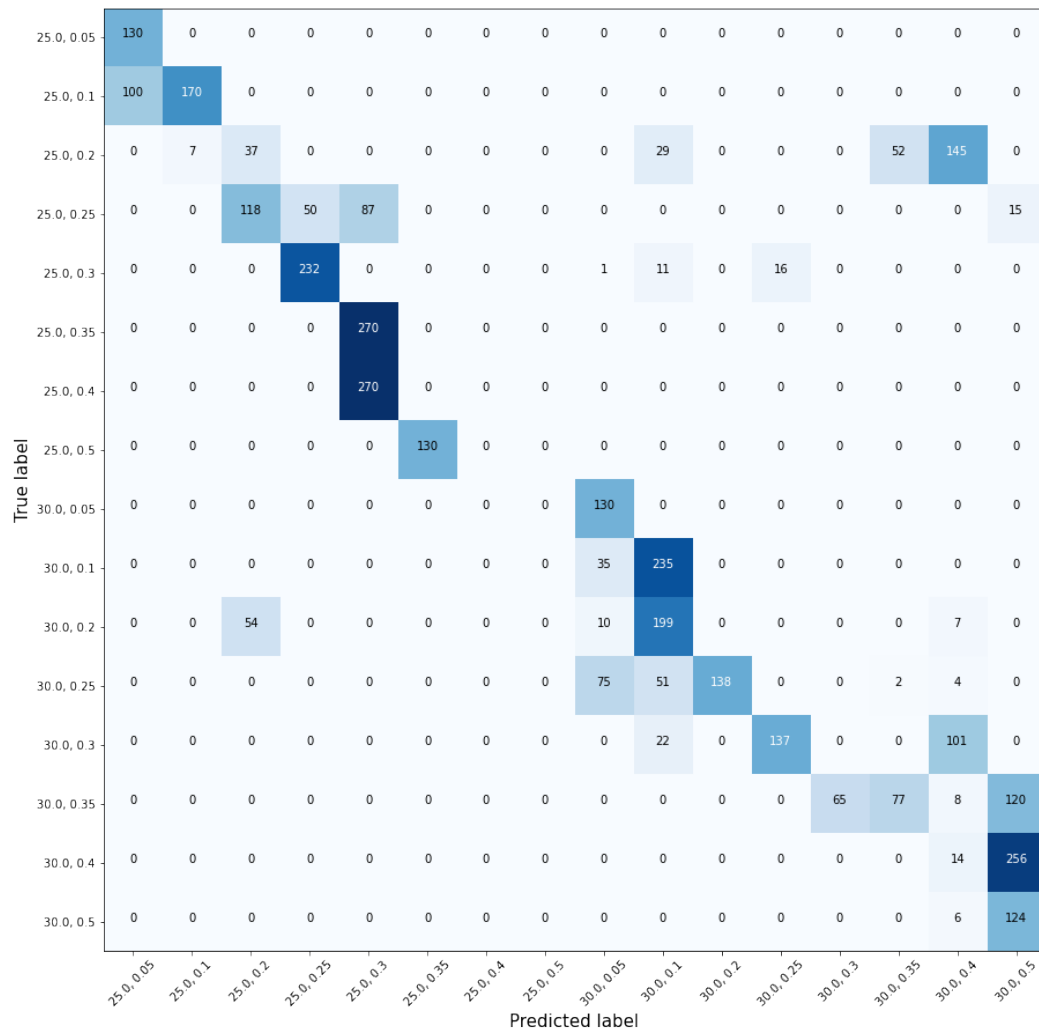


Fig. 14 Confusion matrix for the prediction of faster changing chirp signal on a model trained from sinusoidal + chirp data, with stricter categorization of continuous frequency